DATA SCIENCE & ANALYTICS

MODULE 03 SESSION 14

Improving models: trees, forests and better features

How do we get a better model, and how do we know it is better?

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Sun 25 Oct 2026

BY THE END OF TODAY, YOU CAN

Class objectives

01 Compare models fairly: fixed groups, test once

02 Read a decision tree and predict with it by hand

03 Recognise overfitting from two scores

04 Explain and train forests and boosted trees

05 Engineer a feature; check it for a leak

EVERYONE DO

Step 1.1

Warm-up

Five questions on D13.

Suggested time · 10 minutes

PART 01

A better model, before the theory

Fixed groups for every model, then D11’s line and a prepared boosted model on the same people.

INSTRUCTOR DO

Step 2.1

Today’s question

D11’s insurer, D11’s model, and two questions.

Suggested time · 4 minutes

WHERE D11 LEFT THE INSURER · 1,338 PEOPLE, SIX FEATURES

D11’s model, on rows it had not seen

Score	D11’s line	In words
MAE	≈ 4,200	the average miss, dollars a year
R²	≈ 0.75	explains three-quarters of the variation
misses over 10,000	about 1 in 11	a few people, missed badly

Can we predict yearly charges better, and how would we know that a new model is better?

EVERYONE DO

Step 2.2

The same rows for every model

Three fixed groups, in a view, before any model is trained.

Suggested time · 12 minutes

labs/D14/01-Evr-Same-Rows-Two-Models/README.md

Different columns, different people held aside

	Model A	Model B
features	age, sex, bmi, children, smoker, region	the same six, plus one new column
data_split_method	'RANDOM'	'RANDOM'
the 20% held aside	one set of people	**possibly a different set**

If two models are scored on different people, part of the difference in their scores comes from the people.

DEFINITION

A fair comparison scores every model on the same rows, chosen before any model is trained.

A view that deals the people into three groups

`MOD(ROW_NUMBER() OVER (ORDER BY charges), 5)`: 1, 2, 3, 4, 0, 1, 2 …

```
CREATE OR REPLACE VIEW ml.insurance_groups AS
SELECT *,
       CASE MOD(ROW_NUMBER() OVER (ORDER BY charges), 5)
         WHEN 3 THEN 'validation'
         WHEN 4 THEN 'test'
         ELSE 'train'
       END AS split
FROM ml.insurance;
```

Number the people by charges, then deal by fives

#1	#2	#3	#4	#5	#6	#7	#8	#9	#10
1,122	1,132	1,136	1,136	1,137	1,137	1,141	1,147	1,149	1,163
MOD 5 = 1	MOD 5 = 2	MOD 5 = 3	MOD 5 = 4	MOD 5 = 0	MOD 5 = 1	MOD 5 = 2	MOD 5 = 3	MOD 5 = 4	MOD 5 = 0
train	**train**	**validation**	**test**	**train**	**train**	**train**	**validation**	**test**	**train**

The ten cheapest of 1,338 people, numbered 1 to 10. The pattern repeats to person 1,338.

Remainder 3 → validation, 4 → test, the rest → training. Dealing by the label balances charges on purpose, a teaching choice. Everyone in the class gets the same groups.

BLOCK 2 · CHECK THE DEAL

Three groups that look alike

Group	People	Average charges	Smokers	Its job
train	803	13,267	171	every model learns from it
validation	268	13,346	53	**we compare and choose on it**
test	267	13,204	50	**one final score, at the end**

The test group waits until we have chosen a model. It is used once, for that model.

Practice exams, then one final exam

Group	The student	The model
train	studies practice questions and their answers	learns its weights
validation	sits practice exams, then decides what to revise	we compare models and choose
test	**sits the final exam once, on new questions**	**one final score for the chosen model**

Every choice made on validation suits those rows a little, so the winner's validation score flatters it. Nothing is chosen on test. BigQuery calls validation "evaluation".

EVERYONE DO

Step 2.3

Two prepared models, one table

D11’s line and a boosted-trees model, scored on the same 268 people.

Suggested time · 15 minutes

labs/D14/01-Evr-Same-Rows-Two-Models/README.md

The same six features, our training group

data_split_method = 'NO_SPLIT': we chose the rows; BigQuery holds none aside.

```
CREATE OR REPLACE MODEL ml.ins_linear
OPTIONS (model_type = 'LINEAR_REG',
         input_label_cols = ['charges'],
         data_split_method = 'NO_SPLIT') AS
SELECT age, sex, bmi, children, smoker, region, charges
FROM ml.insurance_groups WHERE split = 'train';
```

Boosted trees, Google’s default settings

`model_type = 'BOOSTED_TREE_REGRESSOR'`: same rows, same features.

```
CREATE OR REPLACE MODEL ml.ins_boosted
OPTIONS (model_type = 'BOOSTED_TREE_REGRESSOR',
         input_label_cols = ['charges'],
         data_split_method = 'NO_SPLIT') AS
SELECT age, sex, bmi, children, smoker, region, charges
FROM ml.insurance_groups WHERE split = 'train';
```

ML.EVALUATE · TWO WAYS TO CALL IT

Which rows did ML.EVALUATE score?

Call	Rows scored
ML.EVALUATE(MODEL m)	the rows held aside when m trained: scored then, saved with the model
ML.EVALUATE(MODEL m, (SELECT …))	**exactly the rows you pass, whichever they are**

Our models use NO_SPLIT, so nothing was held aside: with no SELECT, ML.EVALUATE would score the 803 training people. Block 5 passes the 268 validation people.

Two scores, one table

ML.EVALUATE with a SELECT scores the rows passed: the 268 validation people.

```
SELECT 'linear' AS model, ROUND(mean_absolute_error) AS mae,
       ROUND(r2_score, 2) AS r2
FROM ML.EVALUATE(MODEL ml.ins_linear,
  (SELECT * FROM ml.insurance_groups WHERE split = 'validation'))
UNION ALL
SELECT 'boosted trees', ROUND(mean_absolute_error), ROUND(r2_score, 2)
FROM ML.EVALUATE(MODEL ml.ins_boosted,
  (SELECT * FROM ml.insurance_groups WHERE split = 'validation'));
```

VALIDATION · 268 PEOPLE NO MODEL HAS SEEN

The trees miss by about 1,600 dollars less

Model	MAE (dollars)	RMSE	R²
guess: the training average, 13,267	9,153	12,257	0.00
linear, six features (D11)	4,241	6,262	0.74
boosted trees, defaults	**≈ 2,630**	≈ 5,020	**≈ 0.83**

The line is exact on these rows. Tree numbers can move by about 0.01 in your sandbox. If block 4 fails with a billing message, the Resources README has the fallback.

INSTRUCTOR DO

Step 2.4

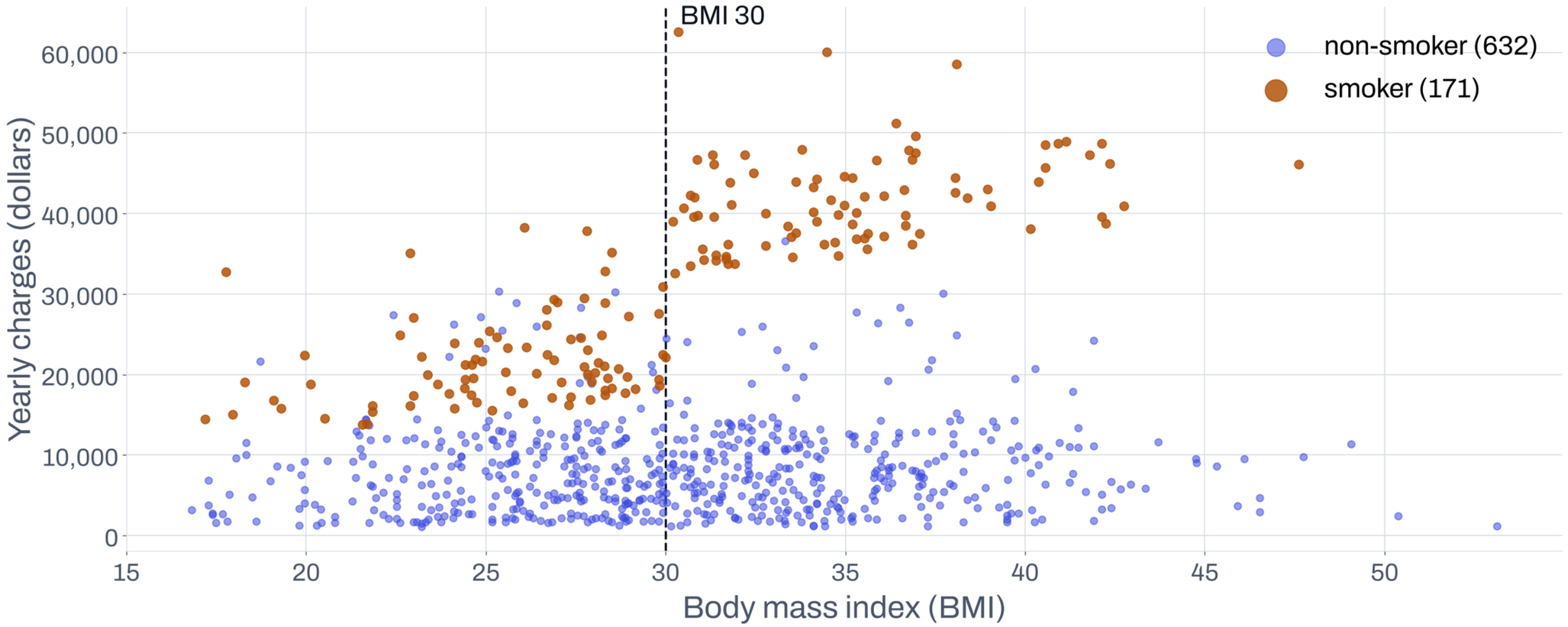
Where the line goes wrong

Charges against BMI, smokers in amber.

Suggested time · 5 minutes

TRAINING ROWS · 803 PEOPLE

Smokers above BMI 30 cost about twice as much



One dot is one person. Indigo: non-smokers. Amber: smokers. Dashed line: BMI 30.

TRAINING ROWS · AVERAGE CHARGES BY GROUP

BMI matters, but only for smokers

	BMI under 30	BMI 30 or more
non-smokers	7,861 (305 people)	8,657 (327)
smokers	21,632 (84)	**41,473 (87)**

A line adds one fixed amount for smoking and one per BMI point, the same for everyone. It cannot say “BMI matters, but only for smokers”.

PART 02

One tree

Read one, use it by hand, see how it chooses its questions, then grow it too deep.

INSTRUCTOR DO

Step 3.1

A decision tree

Yes-or-no questions, one column at a time.

Suggested time · 10 minutes

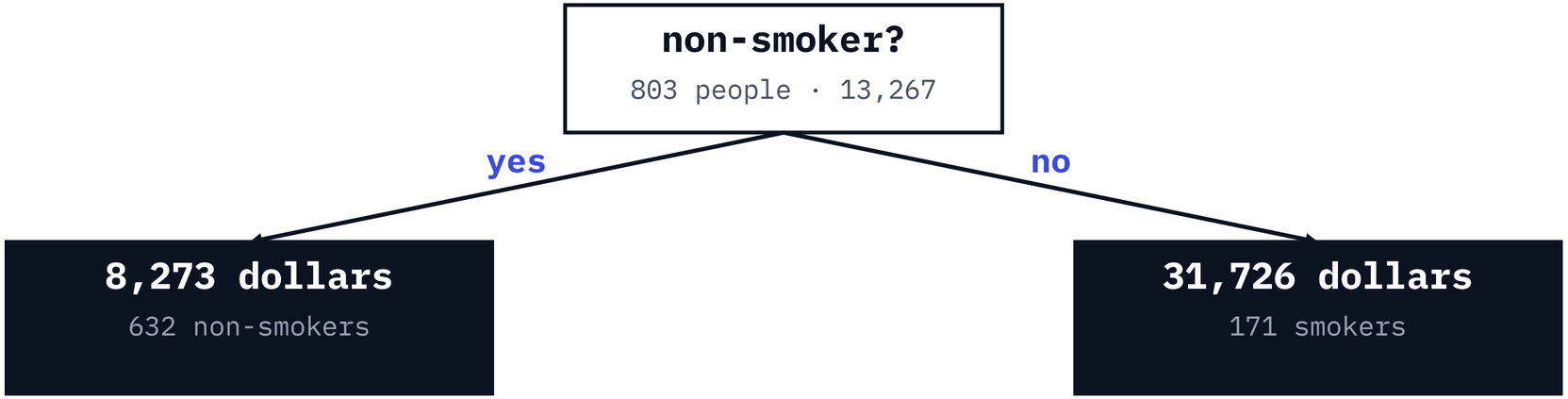
DEFINITION

A decision tree predicts by asking yes / no questions about one column at a time, from the root down to a leaf.

For a number, a leaf holds the average label of the training rows that reached it.

ONE TREE · 803 TRAINING PEOPLE · THE FIRST QUESTION

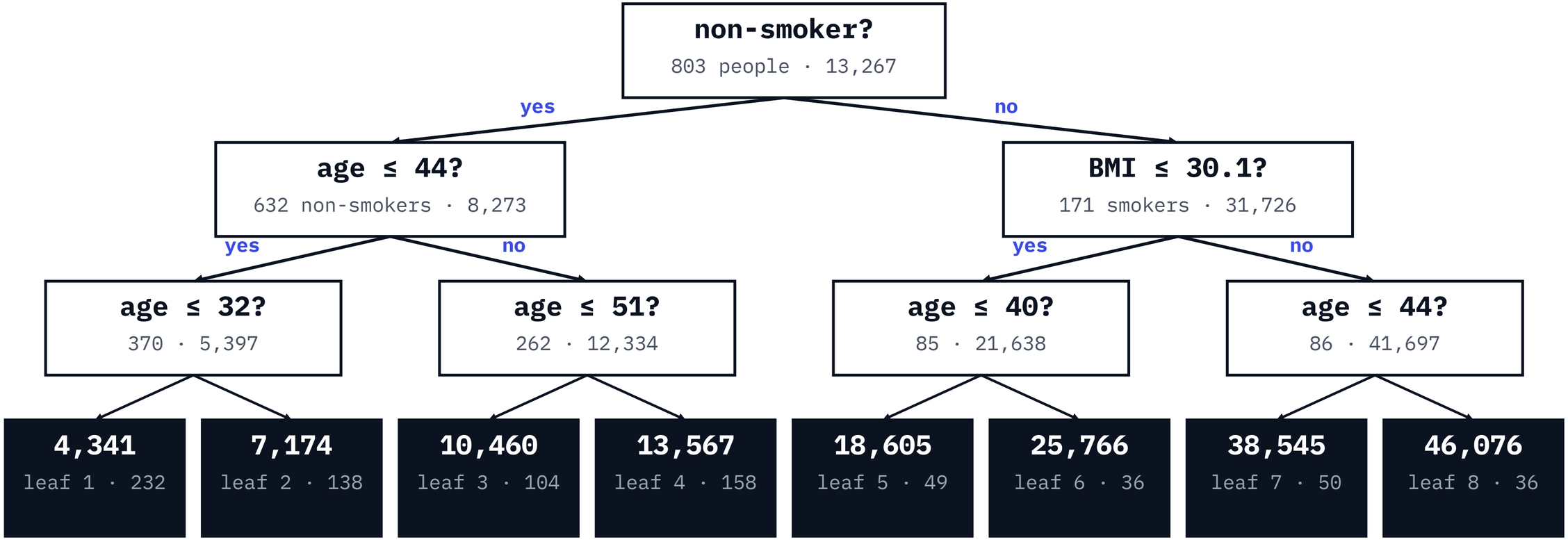
One question: smoker or not?



With one question the tree already gives two answers: 8,273 for every non-smoker, 31,726 for every smoker. Each answer is the average of the training people on that side.

ONE TREE · THREE QUESTIONS DEEP

Eight leaves, eight possible answers

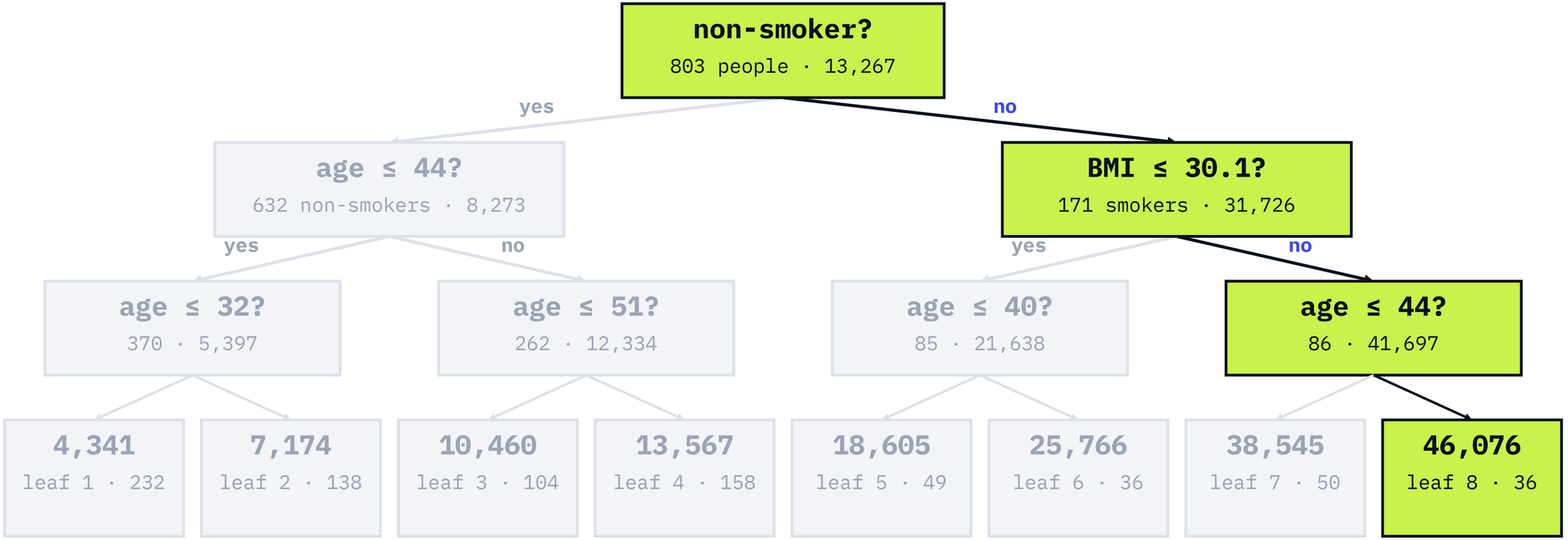


Left branch: yes, right branch: no, at every question. Each box: rows that reached it · their average charges.

Non-smokers: age only. Smokers: BMI first, the pattern the line could not use.

FOLLOW ONE PERSON · E: 57, SMOKER, BMI 31.2

Person E lands in leaf 8: 46,076



Left branch: yes, right branch: no, at every question. Each box: rows that reached it · their average charges.

Non-smoker? no · BMI ≤ 30.1? no · age ≤ 44? no → 46,076. Actual 43,579: a miss of −2,497.

STUDENT DO

Five people through the tree

Route five validation people down the tree, write each prediction and miss, and compare with the line.

Suggested time

10 minutes

`labs/D14/02-Stu-Tree-By-Hand/README.md`

STUDENT DO

Five people, eight leaves

INSTRUCTIONS	Follow each of the five people in the README from the top of the tree to a leaf. Write the leaf, the prediction and the miss (actual – predicted). Average the misses, signs dropped; compare with the line.
HINT	Sex, children and region never change the route.
BONUS	Can this tree ever predict a charge below zero? Why not?

Full steps and Check yourself: labs/D14/02-Stu-Tree-By-Hand/README.md

REVIEW

Time's up. Let's review.

Five people through the tree

`Solved: labs/D14/02-Stu-Tree-By-Hand/Solved/tree_by_hand.md`

THE ANSWERS · FIVE VALIDATION PEOPLE

The tree misses these five by 735 on average

Person	Leaf	Tree	Actual	Tree’s miss	Line’s miss
A · 25, non-smoker	1	4,341	4,392	+51	+1,178
B · 49, non-smoker	3	10,460	10,371	−89	−2,308
C · 60, non-smoker	4	13,567	13,229	−338	−2,034
D · 33, smoker, BMI 24.8	5	18,605	17,905	−700	−10,497
E · 57, smoker, BMI 31.2	8	46,076	43,579	−2,497	+7,512
average miss, signs dropped				**735**	4,706

Eight possible answers, none below zero: each leaf is an average of real charges. Five people are not a test, only a way to see how a tree predicts.

INSTRUCTOR DO

Step 3.4

How a tree chooses its question

Try questions on every column; keep the one that helps most.

Suggested time · 8 minutes

THE FIRST QUESTION · 803 TRAINING PEOPLE

Try the questions; keep the one that shrinks the miss most

Best question on each column	Training RMSE after the split
no question: everyone gets 13,267	12,106
smoker or not	**7,373**
age 41 or under	11,594
BMI 30.2 or under	11,868
children 1 or fewer	12,006
female or male	12,075
southeast or another region	12,091

Then the same search again on each side, using only that side’s rows. That is how the smokers’ side finds BMI and the non-smokers’ side does not.

EVERYONE DO

Step 3.5

One tree, grown too deep

The same tree, three questions deep and fifteen, scored on seen and unseen rows.

Suggested time · 14 minutes

labs/D14/03-Evr-One-Tree-Too-Deep/README.md

A boosted model with one round is one tree

max_iterations = 1, **learn_rate = 1**, **l2_reg = 0**: one tree, kept whole.

```
CREATE OR REPLACE MODEL ml.ins_tree3
OPTIONS (model_type = 'BOOSTED_TREE_REGRESSOR',
         input_label_cols = ['charges'],
         data_split_method = 'NO_SPLIT',
         max_iterations = 1, learn_rate = 1,
         l2_reg = 0, max_tree_depth = 3) AS
SELECT age, sex, bmi, children, smoker, region, charges
FROM ml.insurance_groups WHERE split = 'train';
```

BLOCK 3 · SEEN ROWS AGAINST UNSEEN ROWS

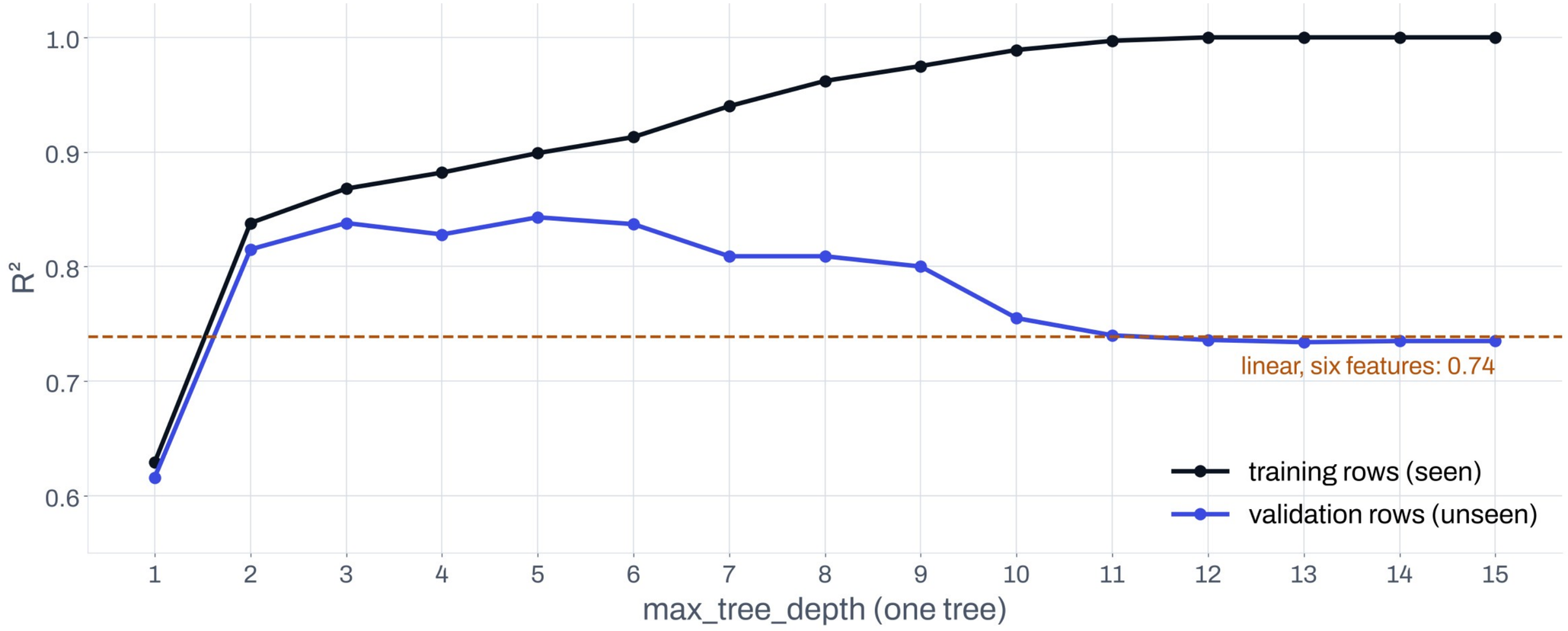
Depth 15: perfect on seen rows, worse on new ones

Tree	Training R^2 (seen)	Validation R^2 (unseen)	Different answers
max_tree_depth = 3	0.87	0.84	8
max_tree_depth = 15	**1.00**	**0.73**	about 770

Depth 15 allows up to 15 questions on each path: nearly every training person got a leaf of their own. On new people its R^2 falls to the line's.

ONE TREE · DEPTH 1 TO 15

Training keeps rising; validation peaks, then falls



Black: R² on the 803 training people. Indigo: on the 268 validation people. Amber: D11’s line on validation.

AGAIN, FROM D11

Overfitting is learning the training rows so closely that the model does worse on rows it has not seen. A large gap between the two scores is the warning sign.

PART 03

Many trees

Side by side and averaged: a forest. One after another: boosting.

INSTRUCTOR DO

Step 4.1

Many trees side by side: bagging and forests

One deep tree is unstable. Many, averaged, are not.

Suggested time · 10 minutes

DEFINITION

A bootstrap sample draws as many rows as the training set, at random, with replacement.

Some rows appear twice, some never. About two thirds of the rows land in each sample.

Rows 4 and 8 twice or more; rows 1, 3 and 6 never

Each draw picks any row, including one already drawn.

rows
1
2
3
4
5
6
7
8

draw 8 with replacement



sample
8
5
2
7
4
4
8
8

DEFINITION

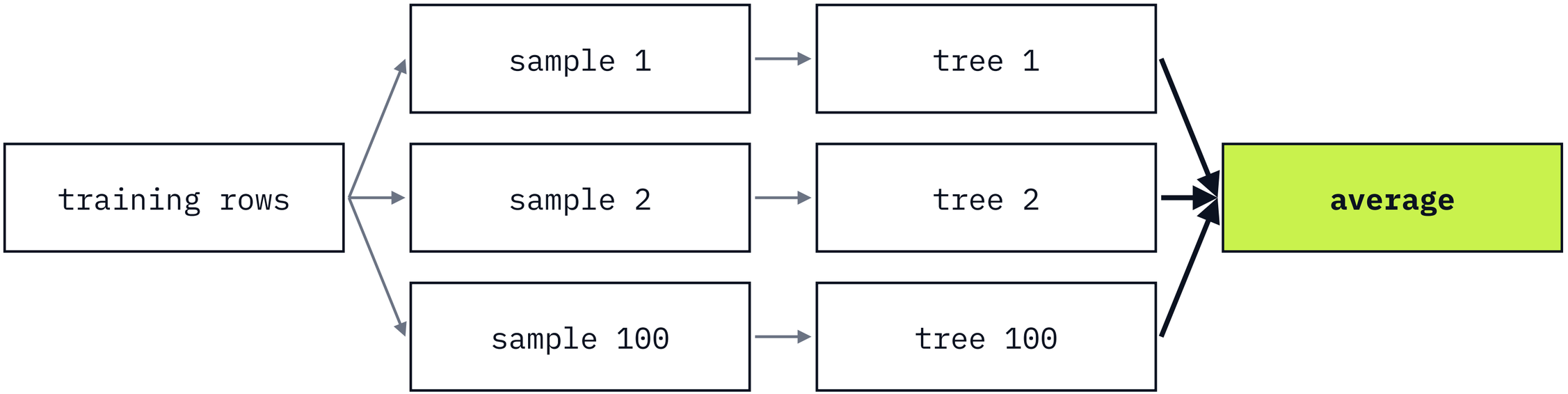
Bagging (bootstrap aggregating) fits one model on each of many bootstrap samples and averages their predictions.

Each tree is unstable in its own way; averaging cancels much of that variance.

BAGGING · THE FLOW

Many samples, many trees, one average

Read left to right: every tree sees a different sample; the prediction is the average of their predictions.



DEFINITION

A random forest is bagging plus a random subset of columns at every split, so the trees make different mistakes.

BigQuery’s forest draws 80% of the columns at each question by default.

RANDOM_FOREST_REGRESSOR · GOOGLE'S DEFAULTS

BigQuery's forest: four settings

Option	Default	What it does
num_parallel_tree	100	trees grown side by side; their predictions are averaged
max_tree_depth	15	each tree may grow deep
subsample	0.8	each tree learns from a random 80% of the training rows
colsample_bynode	0.8	each question chooses among a random 80% of the columns

The bootstrap draws rows with replacement. BigQuery's forest draws its 80% without replacement: each row at most once per tree. Either way, every tree learns from different rows.

INSTRUCTOR DO

Step 4.2

Trees in a sequence: boosting

Small trees one after another, each fixing what is left.

Suggested time · 12 minutes

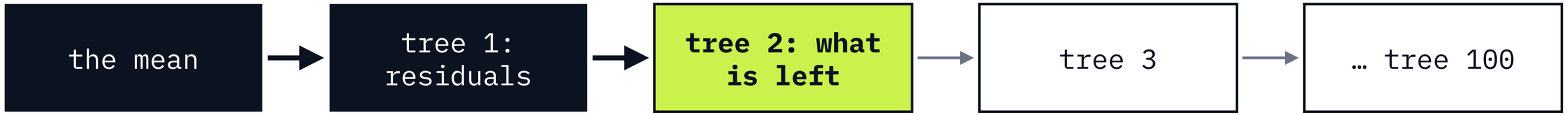
DEFINITION

Boosting fits small trees one after another, each to the errors the model still makes, and adds each correction, scaled down.

A forest averages independent trees. In boosting, each new tree reduces the errors the trees so far still make.

Start from the mean; each tree fits what is left

Lit: tree 2 fits what tree 1 left. Each arrow adds one small tree's correction, times the learning rate.



Start: the average, 31,726, for every smoker



Typical miss (RMSE) 11,522 dollars.

After 1 tree: a step at BMI 30, part of the way



One question, kept at 30% (the learning rate). Typical miss 9,025.

After 3 trees: most of the way



Each tree fits what the trees before it left. Typical miss 6,623.

After 50 trees: a staircase, and a spike



Typical miss 5,097. The spike near BMI 18 chases one person: boosting overfits too, if it runs long enough.

DEFINITION

The learning rate is the share of each tree's correction the model keeps.

Smaller rates need more trees; given enough of them, they often generalize better. BigQuery’s default is 0.3.

BOOSTED_TREE_REGRESSOR · GOOGLE’S DEFAULTS

BigQuery’s boosted trees: five settings

Option	Default	What it does
max_iterations	20	at most 20 rounds: one tree per round
learn_rate	0.3	keep 30% of each tree’s correction
max_tree_depth	6	each tree asks at most six questions on a path
early_stop	TRUE	stop when a round lowers the loss by under 1% (NO_SPLIT: no held-aside rows to watch)
l2_reg	1.0	shrink each leaf’s value, more for small leaves

EVERYONE DO

Step 4.3

A forest and boosted trees on the same rows

Train the forest, then every model of the day in one table.

Suggested time · 15 minutes

labs/D14/04-Evr-Forest-And-Boosted/README.md

A forest, defaults written out

'RANDOM_FOREST_REGRESSOR': 100 trees, averaged.

```
CREATE OR REPLACE MODEL ml.ins_forest
OPTIONS (model_type = 'RANDOM_FOREST_REGRESSOR',
         input_label_cols = ['charges'],
         data_split_method = 'NO_SPLIT',
         num_parallel_tree = 100, max_tree_depth = 15,
         subsample = 0.8, colsample_bynode = 0.8) AS
SELECT age, sex, bmi, children, smoker, region, charges
FROM ml.insurance_groups WHERE split = 'train';
```

BLOCKS 2 AND 3 · EVERY MODEL, THE SAME PEOPLE

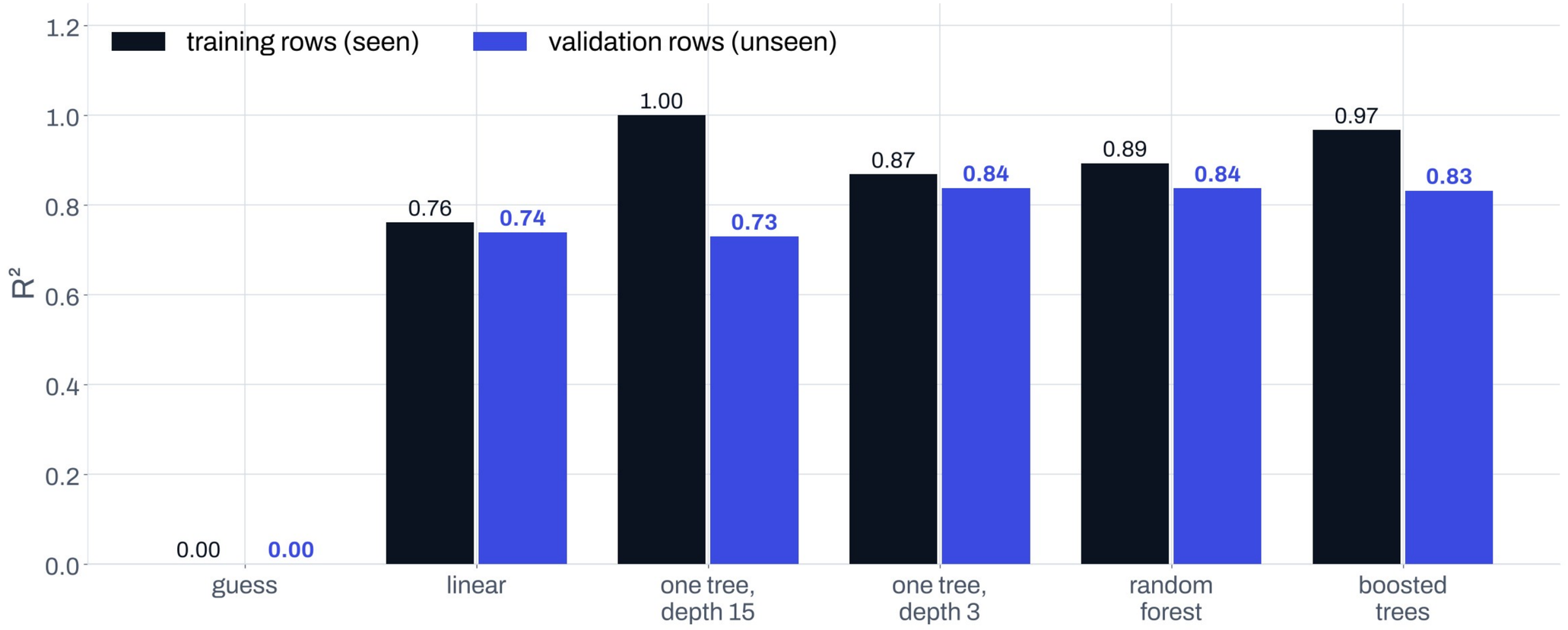
Three tree models land close together

Model	Validation MAE	Validation R^2	Training R^2
random forest	≈ 2,570	**≈ 0.84**	≈ 0.89
one tree, depth 3	≈ 2,810	**≈ 0.84**	≈ 0.87
boosted trees	≈ 2,630	**≈ 0.83**	≈ 0.97
linear, six features	4,241	0.74	0.76
one tree, depth 15	≈ 2,860	≈ 0.73	1.00

The depth-15 tree beats the line on MAE but not on R^2: right for most people, badly wrong for a few, and R^2 squares the misses.

SEEN AGAINST UNSEEN · EVERY MODEL SO FAR

A big gap warns; we choose on validation



Black: R^2 on the training rows. Indigo: on the 268 validation rows.

EVERYONE DO

Step 4.4

Which columns the trees used

How often a column was asked about, and how much it helped.

Suggested time · 7 minutes

labs/D14/04-Evr-Forest-And-Boosted/README.md

Often is not the same as important

importance_weight counts questions; **importance_gain** is the average help per question.

```
SELECT feature, importance_weight,
       ROUND(importance_weight * importance_gain
             / SUM(importance_weight * importance_gain)
               OVER (), 3) AS share_of_gain
FROM ML.FEATURE_IMPORTANCE(MODEL ml.ins_boosted)
ORDER BY share_of_gain DESC;


                          → smoker ≈ 0.68 of all gain, from ≈ 20 questions
```

INSTRUCTOR DO

Step 4.5

Trees for a category

The same trees predict Yes or No: lab 07 tonight.

Suggested time · 6 minutes

DEFINITION

Gini impurity measures how mixed a group is: $1 - p^2 - (1 - p)^2$, where p is the share of Yes.

0: all the same answer. 0.5: a 50 / 50 mix. A classic tree splits to lower it.

REGRESSOR OR CLASSIFIER

Trees for a number, trees for a category

	Predicting a number	Predicting a category
model types	…_REGRESSOR	…_CLASSIFIER
one classic tree: a leaf holds	the average label	the share of Yes
BigQuery (XGBoost): questions lower	the squared miss	the log loss (D12)
BigQuery: the prediction	the trees' values added	their values added, turned into a probability
today	insurance charges	Telco churn, in lab 07

A single tree is for reading; many trees are for predicting

ONE TREE, DEPTH 3 TO 5

Rules a person must read
A pricing rule
No scaling needed

MANY TREES (PART 3)

Accuracy matters
Forests, boosting
Less variance

REVIEW

Time's up. Let's review.

Many trees

Solved: labs/D14/04-Evr-Forest-And-Boosted/Solved/forest_and_boosted.sql

Boosting adds trees in a sequence, each cutting the errors left so far. Try it on any table, judge it on validation, keep a linear model beside it.

Break

30 minutes

After the break: the line learns the trees’ pattern, and we choose a model fairly.

PART 04

Better features, and a fair choice

Hand the line what the trees found, check for a leak, choose, then test once.

INSTRUCTOR DO

Step 5.1

What the tree found

A pattern that holds for one group only.

Suggested time · 6 minutes

Smoker first; BMI only for smokers

	BMI under 30	BMI 30 or more	Difference
non-smokers	7,861	8,657	+796
smokers	21,632	41,473	**+19,841**

When the effect of one column depends on another, it is called an interaction.

DEFINITION

Feature engineering builds a new column from the ones you have, so a model can use a pattern it could not see.

EVERYONE DO

Step 5.2

Hand the line two new features

A switch for smokers with a BMI of 30 or more, and a bend for age.

Suggested time · 14 minutes

labs/D14/05-Evr-Better-Features/README.md

Build the features in a view

`IF(smoker = TRUE AND bmi >= 30, 1, 0)`: 1 for a smoker with a BMI of 30 or more.

```
CREATE OR REPLACE VIEW ml.insurance_features AS
SELECT *,
       IF(smoker = TRUE AND bmi >= 30, 1, 0) AS smoker_obese,
       age * age AS age_sq
FROM ml.insurance_groups;
```

D11’s line, with the trees’ pattern as a column

`smoker_obese, age_sq`: the only change.

```
CREATE OR REPLACE MODEL ml.ins_linear_plus
OPTIONS (
  model_type = 'LINEAR_REG',
  input_label_cols = ['charges'],
  data_split_method = 'NO_SPLIT'
) AS
SELECT age, sex, bmi, children, smoker, region,
       smoker_obese, age_sq, charges
FROM ml.insurance_features
WHERE split = 'train';
```

BLOCK 4 · VALIDATION · THE SAME 268 PEOPLE

Two columns took the line from 0.74 to 0.86

Model	Validation MAE	Validation R²
linear, six features	4,241	0.74
random forest	≈ 2,570	≈ 0.84
boosted trees	≈ 2,630	≈ 0.83
linear + smoker_obese + age_sq	**2,419**	**0.86**

The trees found the pattern. The feature handed it to a model whose weights a pricing manager can read.

BLOCK 5 · WHAT THE NEW LINE LEARNED

Smokers with BMI 30 or more get their own amount

Weight	Value	In words
smoker_obese	**≈ 18,840**	the extra for a smoker once BMI reaches 30, on top of the BMI weight
smoker: true against false	≈ 14,020	the extra cost of smoking below BMI 30 (D11: ≈ 23,800 for all)
age and age_sq	≈ 15 and ≈ 3.1	the yearly step grows with age

Other things equal. A weight shows what the model learned from these rows; it does not prove cause.

INSTRUCTOR DO

Step 5.3

A feature that knows the answer

A colleague’s column, and a score that is too good.

Suggested time · 5 minutes

labs/D14/05-Evr-Better-Features/README.md

BLOCK 6 · THE PLANNED DEAD END

“Monthly charges” from the billing system

charges / 12: built from the label itself.

```
CREATE OR REPLACE MODEL ml.ins_too_good
OPTIONS (
  model_type = 'LINEAR_REG',
  input_label_cols = ['charges'],
  data_split_method = 'NO_SPLIT'
) AS
SELECT age, sex, bmi, children, smoker, region,
       charges / 12 AS monthly_charges, charges
FROM ml.insurance_groups WHERE split = 'train';
```

→ validation R² 1.000 · MAE ≈ 0

Would we know this column when we make the prediction? Ask it of every new feature. If not, it is a leak: perfect on old rows, useless for a new customer.

PARTNER DO

Which model would you ship?

Read the scoreboard and four deals of the rows. Three sentences: which model, why, and how far off for a new customer.

Suggested time

10 minutes

labs/D14/06-Par-Choose-And-Test/README.md

PARTNER DO

Choose on validation, with your partner

INSTRUCTIONS	Use the scoreboard and the four deals in the README. Three sentences for the pricing team: which model, why, and its likely average miss. Post them before the test.
HINT	Look beyond the highest number: steadiness across deals, the training gap, who can read it.
BONUS	Would your choice change if the pricing team only cared about MAE?

Full steps and Check yourself: labs/D14/06-Par-Choose-And-Test/README.md

FOUR DEALS OF THE SAME ROWS · VALIDATION R^2

Scores move by up to 0.04; one model wins every deal

Deal	Linear	Linear + two	Forest	Boosted
1	0.77	**0.89**	0.88	0.86
2	0.75	**0.86**	0.84	0.83
3	0.76	**0.87**	0.86	0.84
4 (today's validation)	0.74	**0.86**	0.84	0.83

Each deal retrains every model and scores a different quarter of the 1,071; test is untouched. This is cross-validation. First every time is evidence, not proof.

EVERYONE DO

Step 5.5

The test, once

The chosen model on 267 people no model and no choice has used.

Suggested time · 8 minutes

labs/D14/06-Par-Choose-And-Test/README.md

One score, for the model we chose

`WHERE split = 'test'`: used once, today, for one model.

```
SELECT ROUND(mean_absolute_error) AS mae,
       ROUND(r2_score, 2) AS r2
FROM ML.EVALUATE(MODEL ml.ins_linear_plus,
  (SELECT * FROM ml.insurance_features WHERE split = 'test'));
```

→ MAE ≈ 2,537 · R² 0.84 · the guess: MAE 9,024

REVIEW

Time's up. Let's review.

Choose, then test once

Solved: labs/D14/06-Par-Choose-And-Test/Solved/choose_and_test.sql

A SAMPLE ANSWER FOR THE PRICING TEAM

Three sentences

#	Sentence
1	Use the linear model with two added features: smoker with a BMI of 30 or more, and age squared.
2	It came first on all four deals of the data, a little ahead of the trees, and its weights can be read and checked.
3	On 267 customers no model had seen, it missed by about 2,500 dollars a year on average, against 4,200 for last week’s model and 9,000 for a flat price.

If only MAE mattered: the same choice. Its validation MAE, 2,419, is also the lowest.

There is no best model, only the best model for a table, a metric and a baseline you can name.

STUDY LIST · IN CLASS AND IN THE LABS

BigQuery ML functions today

In class	What it gives you	Activity
ML.EVALUATE	the scores on the rows you pass: MAE, R^2 and more	01, 03–06
ML.PREDICT	a prediction for each row you pass in	03
ML.FEATURE_IMPORTANCE	trees only: how often and how usefully each feature split	04
ML.FEATURE_INFO	each input column: min, max, mean, NULLs, categories	04
ML.WEIGHTS	the numbers a linear model learned, feature by feature	05

New in the labs	What it gives you	Lab
ML.TRAINING_INFO	the loss after each boosting round	08
ML.TRIAL_INFO	one row per tuning trial: its settings and score	09
ML.BUCKETIZE	a number cut into bands at points you choose	10
ML.QUANTILE_BUCKETIZE	a number cut into bands with about equal counts	10
ML.FEATURE_CROSS	categorical features combined, in pairs	10
ML.POLYNOMIAL_EXPAND	the numbers plus their squares and products	10
ML.METRICS	scores from any table of actual and predicted values (Preview)	Resources

CLOSE

What you can do now

- Compare models fairly; read a tree by hand
- Spot overfitting from two scores
- Explain and train forests and boosted trees
- Build a feature; check it for a leak

NEXT · D15 Forecasting, and explaining what a model predicts.

LAB

AFTER CLASS · SELF-GUIDED

Trees, option by option

TASK

Churn with trees (C1–C3), then three option labs.

FILES

`labs/D14/07-Lab-Trees-For-Churn/`

`labs/D14/08-Lab-Boosted-Tree-Options/`

`labs/D14/09-… and 10-…/`

DONE WHEN

Each lab ends with a table of what you saw.

Optional self-study: lab 11, Nakheel’s spend with trees, where 82 customers are not enough.